Handsfree: Open Source Repos and Resources

Links marked **(verified)** were confirmed in a web search while preparing this list.
Links marked **(check)** are from memory. Open each one and confirm it still exists
and is maintained before you depend on it.

Head / hand tracking and cursor control

- **Project Gameface (Google)** (verified)
  https://github.com/google/project-gameface
  Head-movement and facial-gesture mouse built on MediaPipe. Python/Windows app.
  Best used as a reference for landmark choice, smoothing, and gesture thresholds.
  Announcement: https://blog.google/innovation-and-ai/products/google-project-gameface/

- **Handsfree.js** (verified)
  https://github.com/handsfreejs/handsfree
  Docs: https://handsfree.js.org
  npm: https://www.npmjs.com/package/handsfree
  JavaScript library for face, hand, and pose tracking in the browser, with a
  "Face Pointer" and a browser extension in the repo's /extension folder.
  Apache 2.0 core. Closest match to this problem statement.
  Note: forks of it exist on GitHub; prefer the handsfreejs/handsfree repo.

- **MediaPipe (Face Landmarker / Hand Landmarker, JS)** (check)
  https://github.com/google-ai-edge/mediapipe

Docs:
https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker
Official tracking models that run client-side. Use blendshapes for blink,
smile, and eyebrow gestures.

- **eViacam** (check)
  https://github.com/cmauri/eviacam
  Long-standing open-source webcam head mouse. Useful for calibration ideas.

Cursor smoothing

- **One Euro Filter** (check)
  https://gery.casiez.net/1euro/
  Low-jitter filter for noisy tracking signals. Biggest quality win for cursor feel.

Voice navigation and typing

- **Web Speech API (SpeechRecognition)** (check)
  https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API
  Fastest to integrate. In Chrome it needs an internet connection.
- **Vosk (offline speech recognition)** (check)
  https://github.com/alphacep/vosk-api
  Offline fallback if venue Wi-Fi is unreliable.
- **whisper.cpp (offline speech recognition)** (check)
  https://github.com/ggml-org/whisper.cpp
  More accurate but heavier than Vosk.

Link labels ("click 14") and keyboard-style navigation

- **Vimium** (check)
  https://github.com/philc/vimium
  Chrome extension that overlays labels on links. Reference for your "show numbers" overlay feature.

Browser extension basics

- **Chrome Extensions (Manifest V3) docs** (check)
  https://developer.chrome.com/docs/extensions
  Use the chrome.tabs API for tab commands.